Khwaja Yunus Ali University

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School of Science & Engineering
Department of CSE & EEE
Spring - 2024

Course Outline: Physics – I

Part – A:

1. Course Code:	CSE 0533 1101 (CSE), PHY 0533 1101
2. Course Title:	Physics – I
3. Course Type:	Related
4. Level:	1.1 (1st Year 1st Semester)
5. Academic Session:	Spring - 2024
6. Course Instructor:	Md. Matiur Rahman
7. Prerequisites:	N/A
8. Credit Value:	3.00
9. Contact hours/week:	3.00 hours
10. Total Marks:	100
11. Rationale of the Course:	This course is to give students a broad understanding of physics and a foundation in problem solving and critical thinking. Topics covered in this course include electricity, magnetism and waves & oscillation.

12. Course Objectives

The principles of physics are the basis of all modern technology, and understanding the concepts of physics and knowing how to solve physics problems is a key indicator of success in advanced study in all technical fields, including biology, medicine, and the health sciences. The purpose this course is to give students a broad understanding of physics and a foundation in problem solving and critical thinking. Topics covered in this course include electricity, magnetism and waves & oscillation.

13. Course Learning Outcomes (CLOs) and its mapping with PLOs:

CLOs	After successful completing this course, students will be able to.	PLO Mapping
CLO1	Define the basic terms regarding electricity, magnetism and waves & oscillation.	PLO1
CLO2	Apply the critical thinking and problem-solving skills to solve the variety of basic problems in electricity, wave motion, magnetism.	PLO2 & PLO9
CLO3	Demonstrate the fundamental physics principles using modern tools.	PLO5
CLO4	Analyze and describe the natural phenomenon which are observed in practical situation in everyday life.	PLO4 & PLO7
CLO5	Create ideas to do the research work for the advancement of technological world.	PLO3 & PLO11,12
CLO6	Apply ethical responsibilities and professional conduct.	PLO8

Part – B:

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week	Topic	Teaching Strategy	Assessment Technique	Alignment with CLOs
01	Concept of electric charge, conductors and insulators, permittivity of a medium, Coulomb's law	Interactive discussion / Group activity / Discussion / Presentation / Laboratory Work	Quiz / Class test / Assignment / Presentation / Midterm / Final exam.	CLO1, CLO2 & CLO4
02	The electric field, lines of force, dipole in an electric field, electric flux, Gauss' law			CLO2 & CLO3
03	Electric potential, relation between electric potential and electric field			CLO2 & CLO3
04	Capacitance, calculation of capacitance, different types of capacitors, capacitors with dielectric			CLO1, CLO2 & CLO3
05	Energy storage in an electric field, charging and discharging of a capacitor, time constant			CLO2, CLO3 & CLO5
06	Permeability of a medium, the magnetic field, Biot-Savart law			CLO1 & CLO3
07	Ampere's law, magnetic force on a current, magnetic lines of induction			CLO2 & CLO3
08	Force between two parallel current carrying conductors, Hall Effects			CLO2, CLO3 & CLO4
09	Faraday's law, Lenz's law, self and mutual induction			CLO2, CLO3 & CLO4
10	Transformer and transient response in LR circuit. Physical			CLO2 & CLO4
11	Differential equation of a Simple Harmonic Oscillator, Total energy & average energy, Combination of simple harmonic oscillation			CLO1 & CLO2
12	Differential equation of a progressive wave, Power & intensity of wave motion, Stationary wave, Group velocity & Phase.			CLO2 & CLO3
13	Combination of simple harmonic oscillation, Spring-mass system, Two-body oscillation. Reduced mass			CLO2, CLO3 & CLO4
14	Calculation of time period of torsional pendulum, Damped oscillation, Determination of damping co-efficient, Forced oscillation, Resonance, Lissajous figures			CLO4, CLO5 & CLO6

Part – C:

15. Assessment and Evaluation

1. **Assessment and Evaluation**

1. Assessment Strategy

Students are evaluated out of total 100 Marks. There are different types of assessment tools. In **the Mid Term** examination, total time **is 1.5 hours** where the marks is **30**. In the **Final Examination**, total marks is **40** and allotted time is **2 Hours**. The detail with **CLOs-Assessment Mapping** is given below.

CLOs	Assessment Tools (100%)					
	20%			30%	40%	10%
	Class Test	Quizzes	Assignments	Midterm Exam.	Final Exam.	Attendance
CLO1	√	√		√	√	√
CLO2	√	√		√	√	√
CLO3			√	√	√	√
CLO4			√	√	√	√
CLO5				√	√	√
CLO6			√			√

2. Evaluation System (Grading System)

Numerical Number	Letter Grade		Grade Point
80 % and above	A+	(A Plus)	4.00
75 % to less than 80 %	A	(A Regular)	3.75
70 % to less than 75 %	A-	(A Minus)	3.50
65 % to less than 70 %	B+	(B Plus)	3.25
60 % to less than 65 %	B	(B Regular)	3.00
55 % to less than 60 %	B-	(B Minus)	2.75
50 % to less than 55 %	C+	(C Plus)	2.50
45 % to less than 50 %	C	(C Regular)	2.25
40 % to less than 45 %	D		2.00
Less than 40 %	F		0.00

2. **Marks Distribution**

a) Continuous assessment (30 Marks)

Bloom's Category Marks (out of 30)	Class Tests (07)	Assignments (07)	Quizzes (06)	Curricular / Co-Curricular Activities (10)
Remember	02			Attendance
Understand	02			
Apply	03			
Analyze		02		
Evaluate		02	03	
Create		03	03	
Total	07	07	06	10

b) Summative: Semester (Midterm 30 marks and Final exam 40 marks) Total: 70 Marks.

Bloom's Category	Midterm (30)	Final (40)	Others Evaluation
Remember	5	8	10
Understand	5	8	
Apply	5	6	10
Analyze	5	6	
Evaluate	5	6	10
Create	5	6	
Total	30	40	

3. **Make up procedures:**

For weak students, we take extra class before their midterm and final examinations. Beside this, we counseling with them and solve their problem as soon as possible.

Part – D:

16. Learning Materials:

1) Recommended Reading

- Physics for Engineers (Part – 1 & Part – 2) – Gias Uddin Ahmed
- Halliday Resnick - Fundamental of Physics, 2nd edition
- Brijlal - Heat and Thermodynamics - 1st edition
- Tofazzal Hossain - A Text Book of Heat, 2nd edition
- Beiser - Concept of Modern Physics, 6th edition

2) Supplementary Reading

- Web Source Link
- Journal papers
- Physics for Engineers – Jiaul Ahsan

3) Others:

- Lecture Outlines
- PowerPoint File
- Webpage